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PROJECT COMPLETION CERTIFICATE

This is to certify that Miss. **Chitranshi Yadav** of **BSc Computer Science Semester-IV** has completed the project titled **Railway Track Signal Network** in the Subject of **Computer Network** during the academic year 2025-26 under the supervision and guidance of Prof. **Satvika Guntha** being the partial requirement for the fulfillment of the curriculum of Degree of Bachelor of Science in Computer Science, University of Mumbai.

Name & Signature of Guide Name & Signature of external

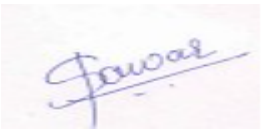
Evaluation sheet for continuous assessment with rubrics

Class : Bachelor of Science (Computer Science)
Subject : Computer Network

Details about the continuous Assessment 2/Project Work
Name : Chitranshi Yadav
Rollno: 5275
Class : SYCS A
Name of Evaluator: Prof. Satvika Guntha

Please circle appropriate score

Grading Criteria	Fair	Good	Excellent	Total
Introduction/ Description of the Case	1	2	3	/3
SWOT Analysis of the company used for case analysis pertaining to the case (strength of CA2 topic: e.g main important feature of CA1, Weakness: limitations of the project, Opportunities: in carrier in the future, Threat: obstacles that can cause failure to project CA2	3	4	5	/5
Learnings from the case	2	3	4	/4
Delivery/presentation skills	1	2	3	/3
Total				/15



Coordinator : Prof. Shubhangi Pawar

Mahatma Education Society's

PILLAI COLLEGE OF ARTS,COMMERCE AND SCIENCE

(Autonomous)

Re-accredited "A" Grade by NAAC (3 rd Cycle)

PROJECT REPORT ON

Railway Track Signal Network

IN PARTIAL FULFILLMENT OF

BACHELOR OF COMPUTER SCIENCE

(SECOND YEAR)

PROJECT GUIDED BY

PROF. SATVIKA GUNTHA

PROJECT SUBMITTED BY

CHITRANSHI YADAV

ROLLNO: 5275

CLASS: SYCS (A)

Introduction of Railway Track Signal Network Project

The Railway Track Signal Topology project is designed to create a digital and network-based system for managing railway signals, ensuring safe and efficient train operations.This system divides the railway tracks into multiple blocks, each controlled by signals connected through a network of routers and switches. The central controller monitors the status of every track and signal, allowing trains to move without collisions and optimizing train traffic.

Using network protocols such as OSPF, the system efficiently routes information between the control center and track devices. This technology is widely used in modern railway systems worldwide to enhance safety, reduce delays, and support real-time monitoring of train movements.
Therefore, the Railway Track Signal Topology project not only demonstrates the practical application of networking in railway management but also highlights the importance of technology in improving safety and operational efficiency in transportation systems.

The Railway Track Signal Network project is a communication network designed for connecting different railway stations using computer networking.
This project uses:

- Ring topology+ Star topology
- 6 routers
- 6 switches
- 12 PCs (2 PCs per station)
- OSPF dynamic routing protocol

The main aim of this project is to provide **reliable and continuous communication** between railway stations for signal control, message transfer and monitoring.
If any one link fails, the data can travel in the opposite direction of the ring, so communication does not stop.This network is similar to real railway optical fiber backbone used in modern railways.

Domain Name:

→ Railway Domain

The project is in the **Railway Domain**, focusing on safe and efficient train operations using networked signals.

Used Topology:

→ Ring Topology+ Star Topology

a) Ring Topology

- **Description:** All devices (signals, sensors, switches) are connected in a closed loop. Data flows in one direction (or both directions if dual ring).

- **Why used:**
    1. Provides **continuous communication** between all signals.
    2. Easy to detect faults in the loop.
    3. Ensures **no single point of failure** if a dual ring is used.
    4. **Ring** ensures **redundancy and reliability** for critical signals.

b) Star Topology

- **Description:** All devices connect directly to a **central controller** (like a server or main control room).

- **Why used:**
    1. Centralized monitoring makes **management simple**.
    2. Failure of one signal does **not affect others**.
    3. Ideal for **smaller track sections** or controlled areas.
    4. **Star** allows **easy monitoring and control** from the central controller.

Used Components in network:

1) Router(2911)
2) Switch(2960)
3) Pc's
4) Copper straight cable
5) Copper Cross over cable

1) Router – Cisco 2911

Role: Connects different track segments and manages data flow between the signals, switches, and central controller.

Why used:

- Handles multiple connections efficiently.

- Supports routing protocols like OSPF for managing communication.
- Ensures data between signals and control center travels reliably.

2) Switch – Cisco 2960
Role: Connects multiple devices (signals, PCs, sensors) within the same track section.
Why used:

- Provides fast, local data exchange.
- Manages network traffic efficiently.
- Helps in forming Star topology segments within the network.

3) **PCs**

Role: Act as central controllers or monitoring stations.
Why used:

- Operators can monitor the status of all tracks and signals.
- Can configure routers, switches, and check network status.

4)Copper Straight-Through Cable
Role: Connects **different devices of different types**, e.g., PC → Switch or Router → Switch.
Why used:

- Standard cable for connecting end devices to network devices.
- Ensures proper data transfer between PCs, switches, and routers.

5) Copper Crossover Cable
Role: Connects the same **type of devices**, e.g., Switch → Switch or Router → Router.
Why used:

- Allows direct communication between two devices of the same type.
- Used in ring connections for redundancy in your Ring topology.

Working:

1. **PC to Switch Communication (Star Topology)**
    - Each PC connects to its local switch (e.g., PC0 → Switch0).
    - PCs send train position or signal data to the switch.
    - Switch forwards data to the connected router in the same Star segment.

2. **Router Communication (Ring Topology)**
    - Routers (2911 series) are connected in a ring loop.
    - Data from one router can travel clockwise or counter-clockwise to reach the target router.
    - Example: Data from Switch0 → Router0 → Router1 → Router2 to reach PC4.
    - This ensures fault tolerance: if one router or link fails, data can take the other path.

3. **Central Monitoring & Control**
    - PCs in critical locations (e.g., PC2, PC3, PC9, PC10) can act as central monitoring stations.
    - They receive data from all track sections and control signals automatically.

4. **Data Flow Example**
    - Train enters block near PC0: Track sensor sends data to Switch0 → Router0 → Router2 → Switch2 → PC4 (monitoring station).
    - Signal updates: Controller decides whether to turn the signal red, yellow, or green.
    - Communication happens in real-time via routers and switches.

5. **Redundancy & Reliability**
    - Ring connection between routers ensures continuous communication even if a link goes down.
    - Star topology for PCs ensures centralized monitoring and easy addition of new devices.

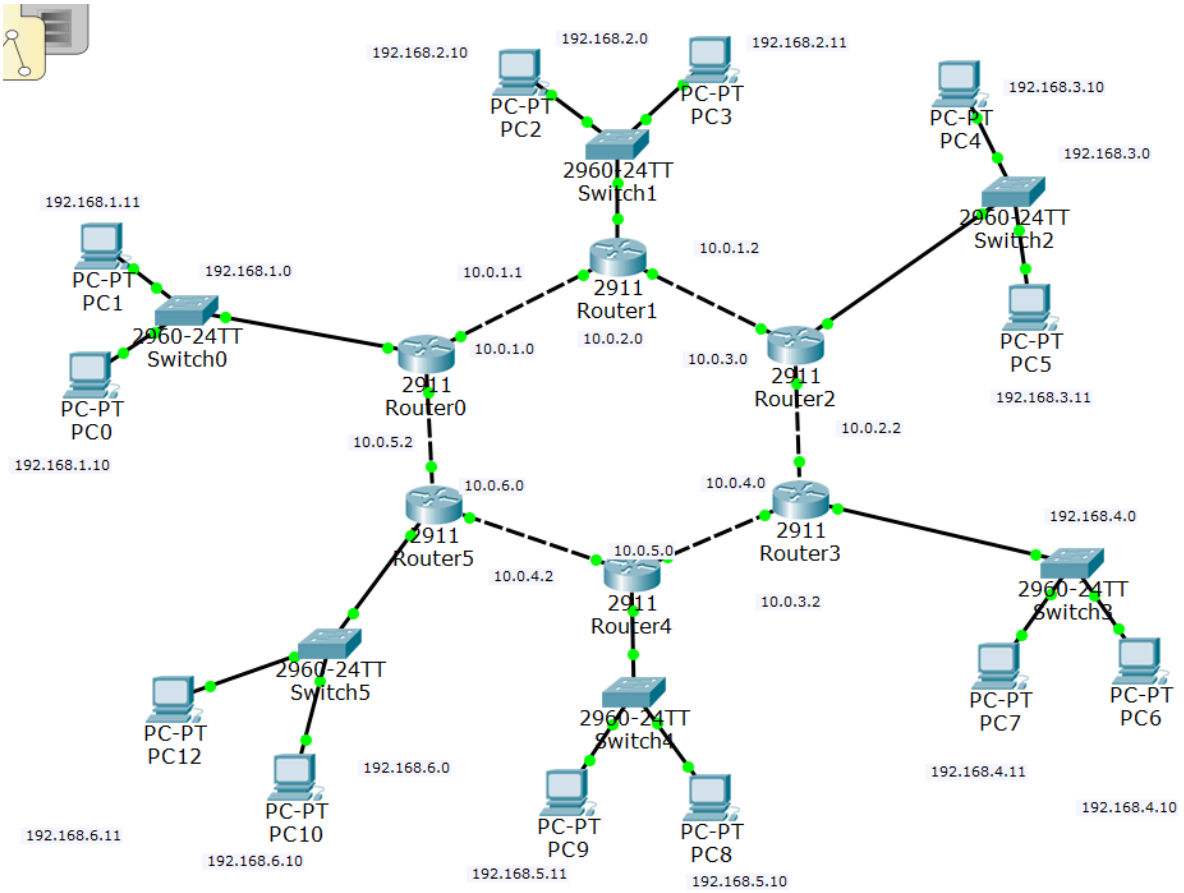
Objectives Of The Project

1. To design a **Railway Track Signal Network** using computer networking concepts.
2. To implement **Ring Topology** for connecting multiple railway stations.
3. To configure **6 routers, 6 switches and 12 PCs** for inter-station communication.
4. To use **OSPF routing protocol** for automatic and dynamic routing.
5. To provide **continuous and reliable communication** between stations.
6. To achieve **fault tolerance**, the network works even if one link fails.
7. To enable data transfer such as:
    - signal information
    - control messages

- station monitoring.

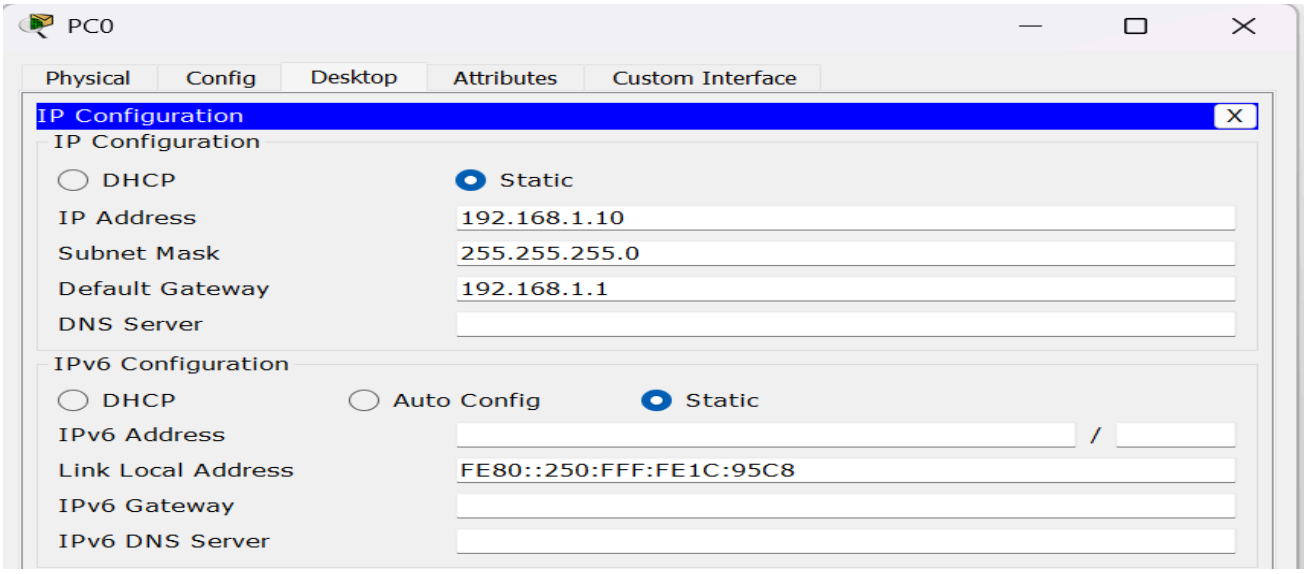
8. To create a network model similar to a real **railway communication system**.
9. To understand practical configuration of:
    - IP addressing
    - routing
    - device connectivity.

NETWORK:

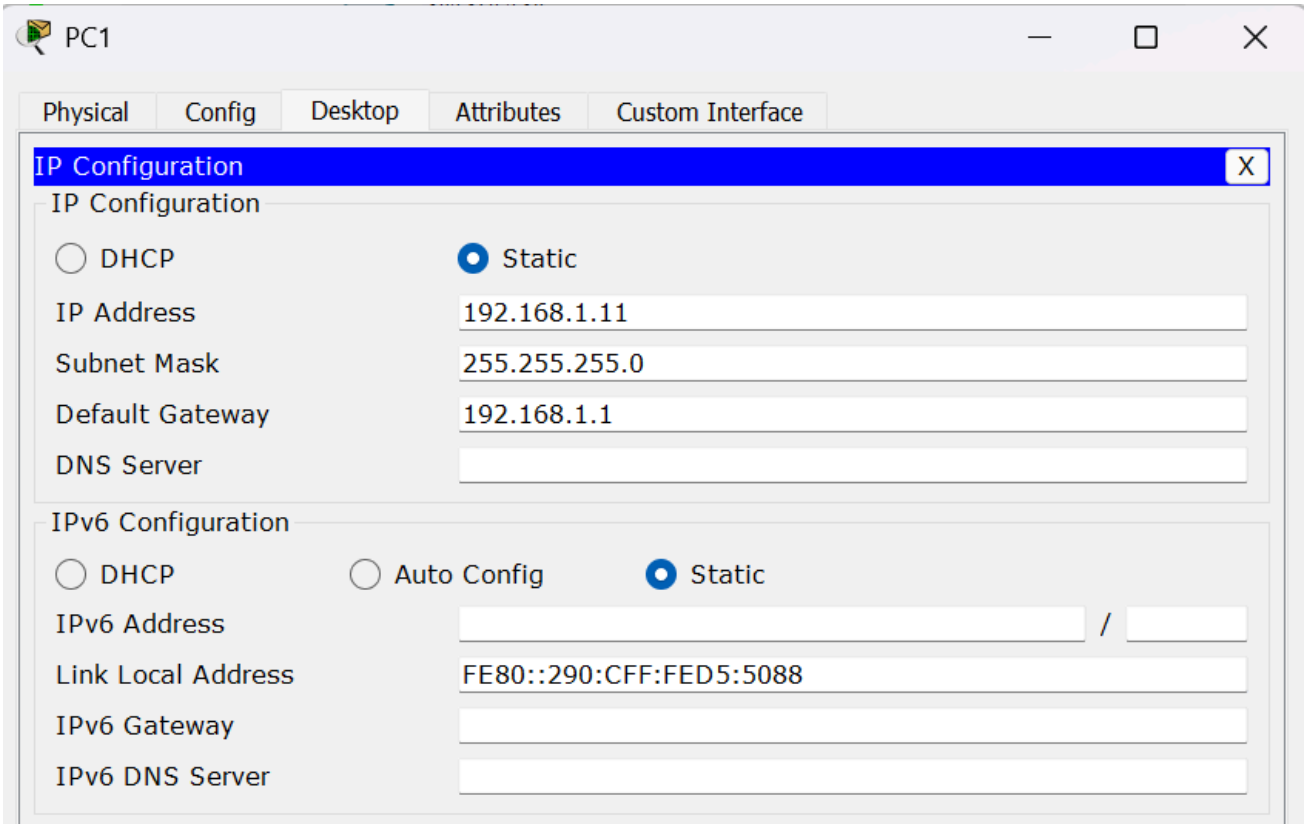


PC IP CONFIG

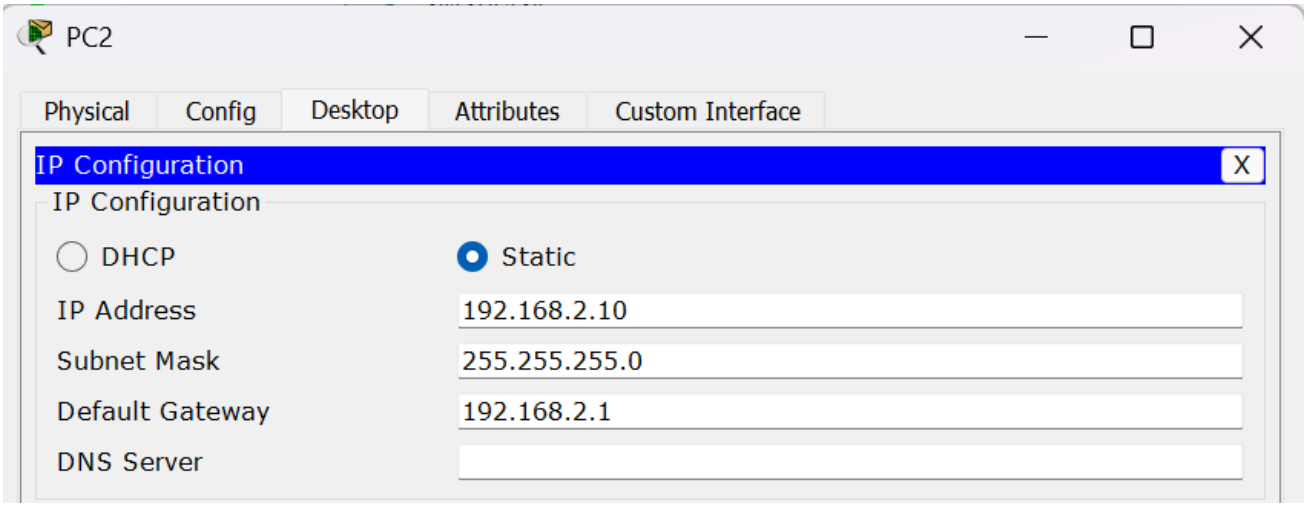
1) IP configuration of PC0



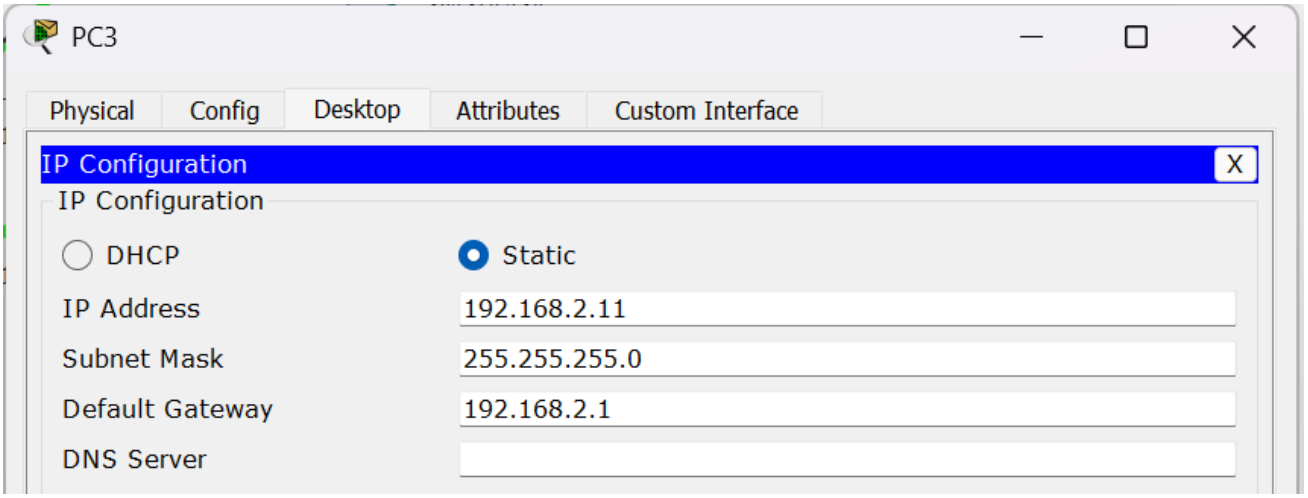
2) IP configuration of PC1



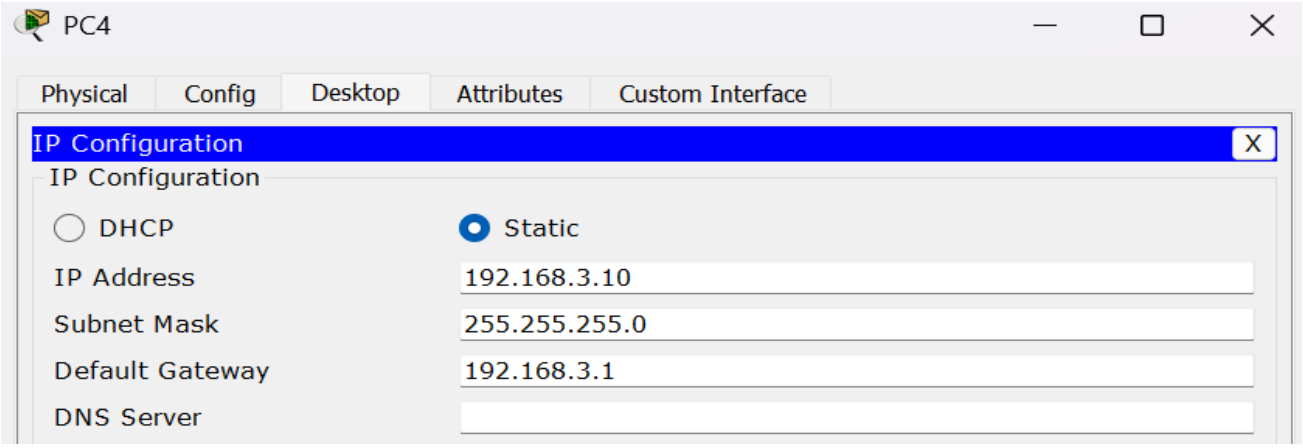
3) IP configuration of PC2



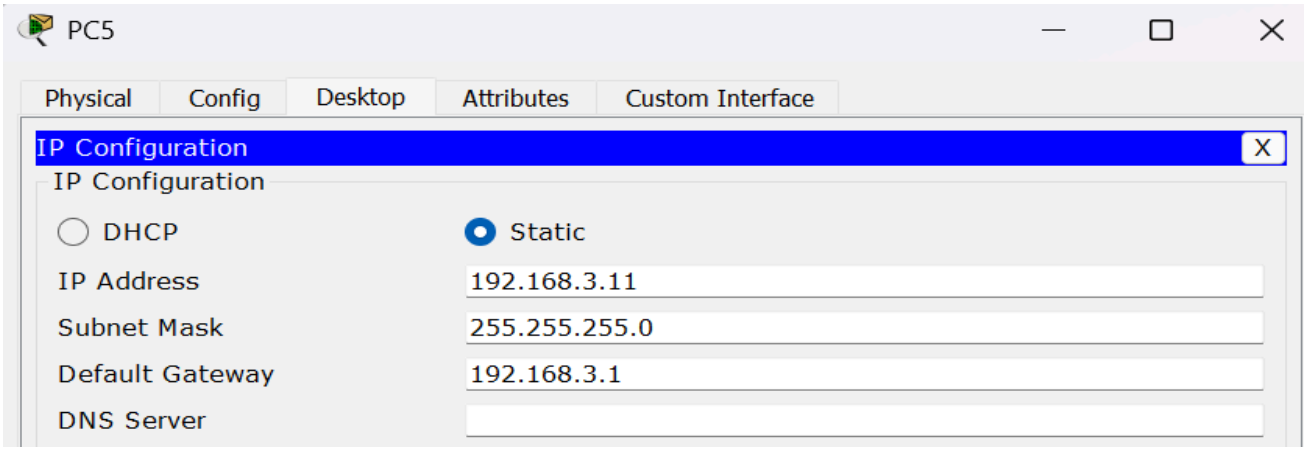
4) IP configuration of PC3



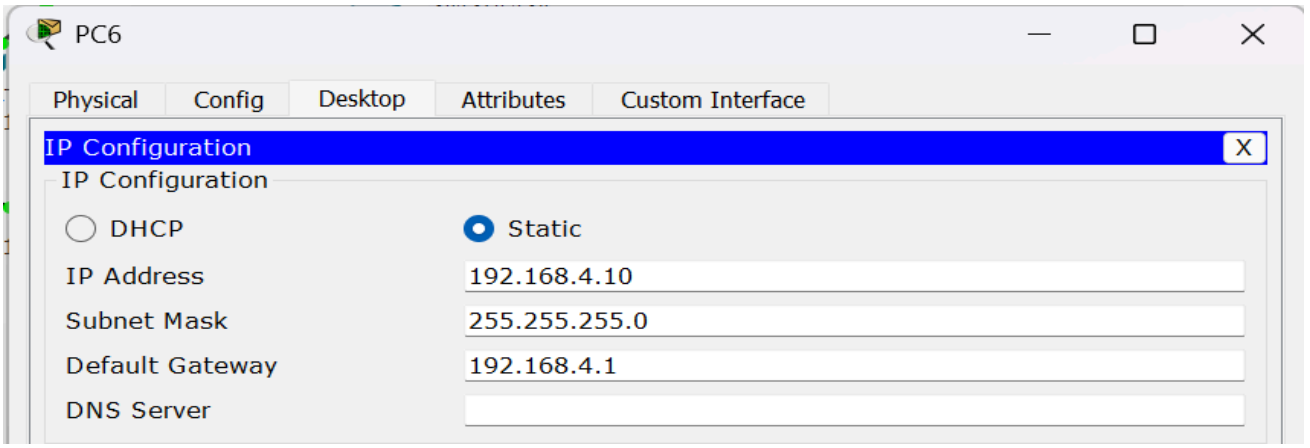
5) IP configuration of PC4



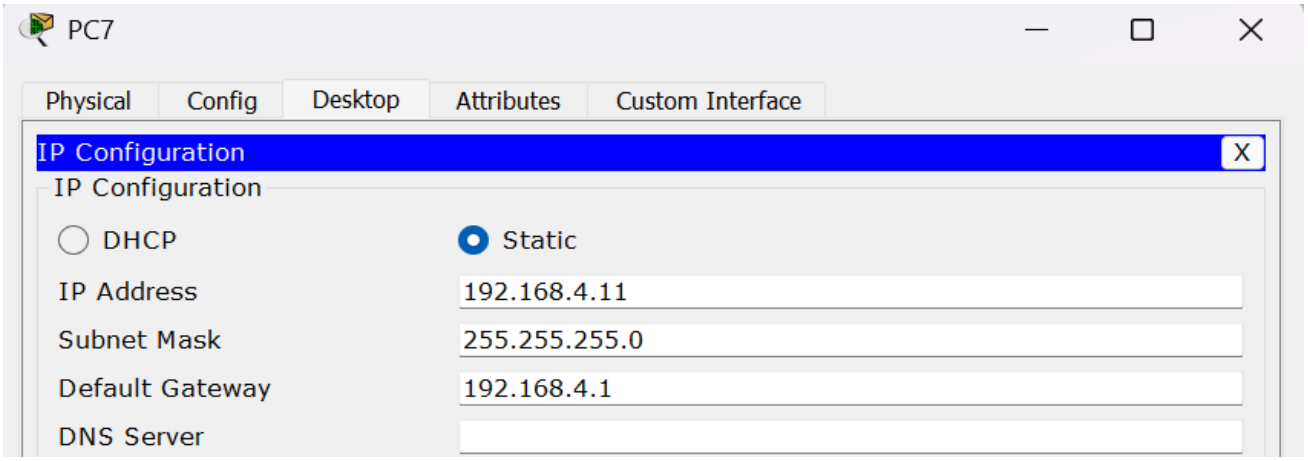
6) IP configuration of PC5



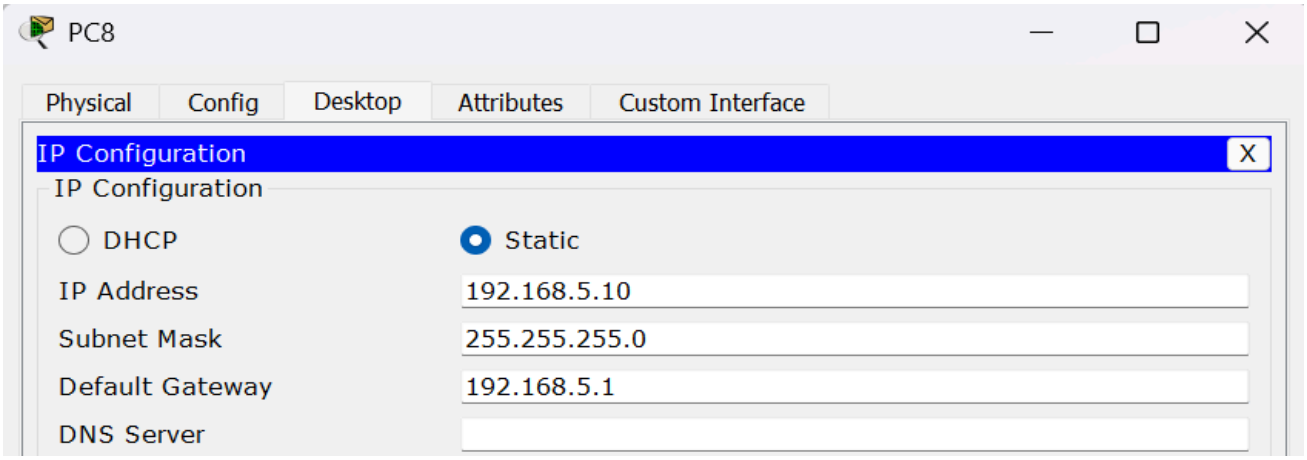
7) IP configuration of PC6



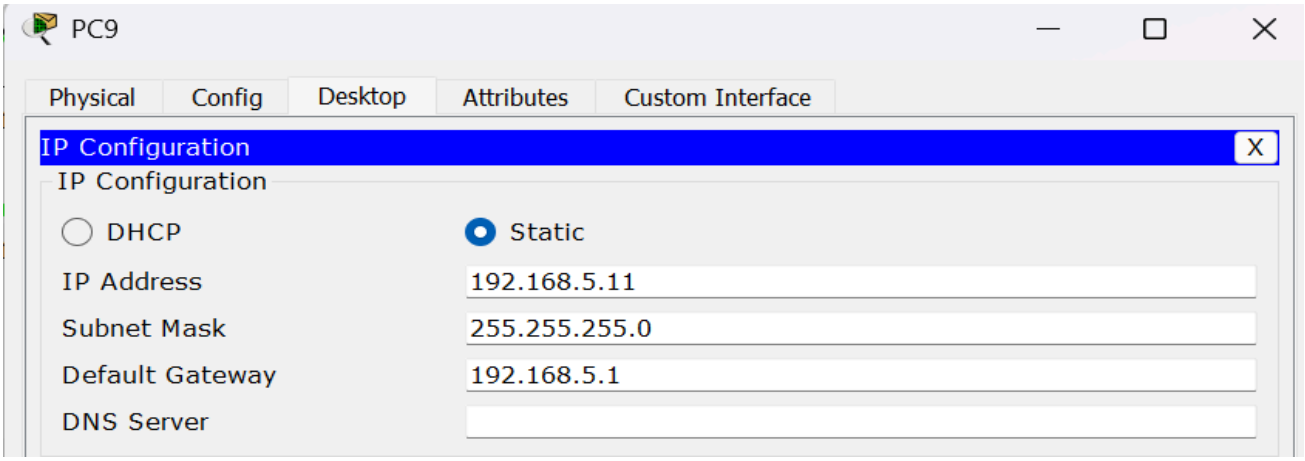
8) IP configuration of PC7



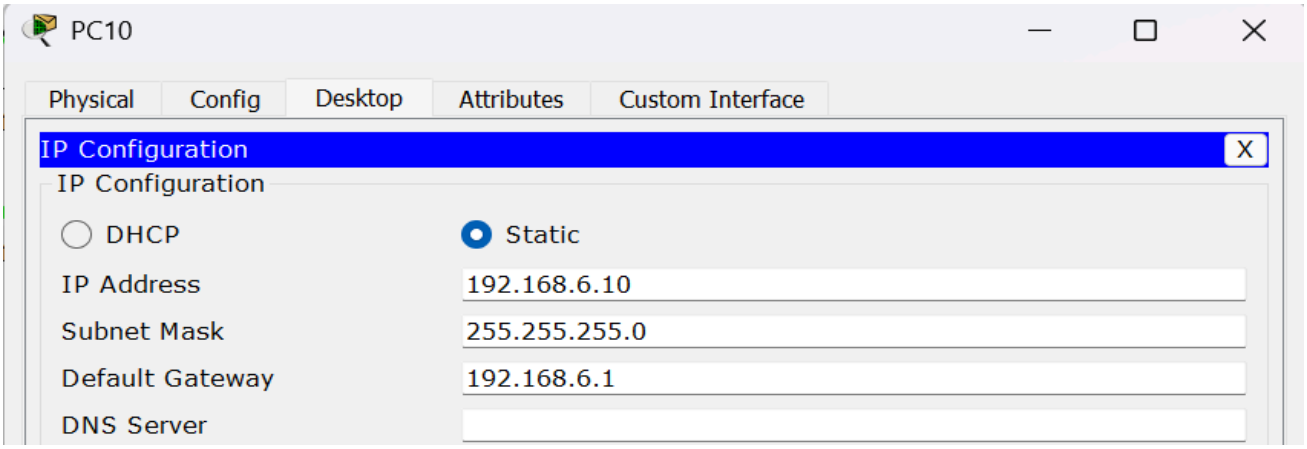
9) IP configuration of PC8



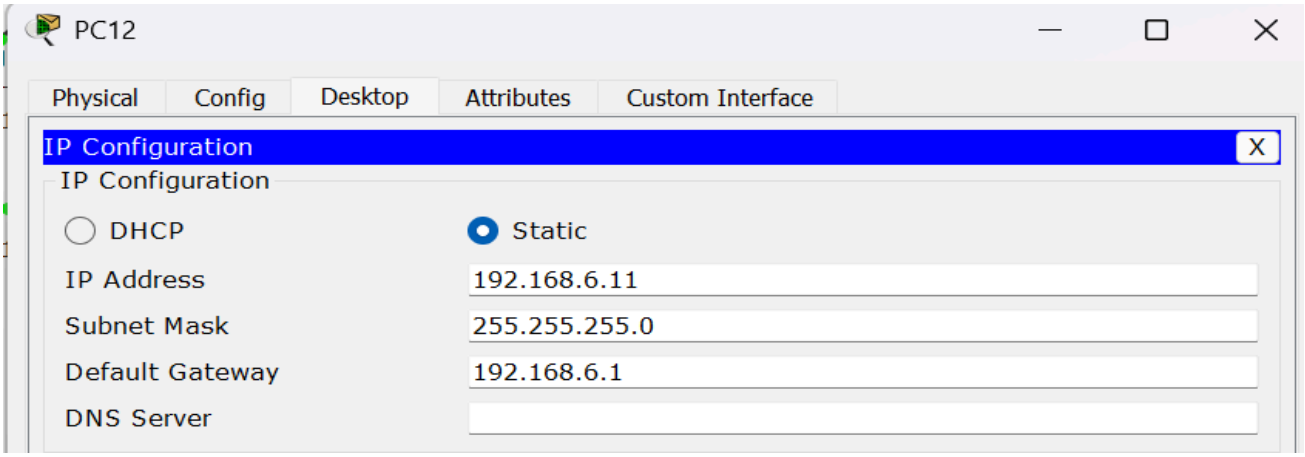
10)IP configuration of PC9



11)IP configuration of PC10



12)IP configuration of PC12



Router ospf configuration

Router0

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
CNTL/Z.
Router(config)#router ospf 1
Router(config-router)#network 10.0.1.0 0.0.0.3 area 0
Router(config-router)#network 10.0.2.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area
0
Router(config-router)#exit
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.1.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.6.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 10.0.1.0 0.0.0.3 area 0
Router(config-router)#network 10.0.6.0 0.0.0.3 area 0
Router(config-router)#network 192.168.1.0 0.0.0.255 area
0
Router(config-router)#exit
```

Copy | Paste

Top

Router1

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.1.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.2.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 10.0.1.0 0.0.0.3 area 0
Router(config-router)#network 10.0.2.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area
0
Router(config-router)#exit
Router(config)#
```

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Top

Router2

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.2.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.3.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 10.0.2.0 0.0.0.3 area 0
Router(config-router)#network 10.0.3.0 0.0.0.3 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area
0
Router(config-router)#exit
Router(config)#
```

Copy | Paste

Router3

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.3.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.4.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.4.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 10.0.3.0 0.0.0.3 area 0
Router(config-router)#network 10.0.4.0 0.0.0.3 area 0
Router(config-router)#network 192.168.4.0 0.0.0.255 area
0
Router(config-router)#exit
Router(config)#
Router(config)#
```

Copy | Paste

Router4

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.4.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.5.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.5.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 10.0.4.0 0.0.0.3 area 0
Router(config-router)#network 10.0.5.0 0.0.0.3 area 0
Router(config-router)#network 192.168.5.0 0.0.0.255 area
0
Router(config-router)#exit
Router(config)#
```

Copy | Paste

Router5

Physical | Config | CLI | Attributes

IOS Command Line Interface

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line.  End with
CNTL/Z.
Router(config)#
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.5.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/1
Router(config-if)#ip address 10.0.6.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#interface g0/2
Router(config-if)#ip address 192.168.6.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#
Router(config)#router ospf 1
Router(config-router)#network 10.0.5.0 0.0.0.3 area 0
Router(config-router)#network 10.0.6.0 0.0.0.3 area 0
Router(config-router)#network 192.168.6.0 0.0.0.255 area
0
Router(config-router)#exit
Router(config)#
```

Copy | Paste

Final output as ping:

PC3

Physical | Config | Desktop | Attributes | Custom Interface

Command Prompt

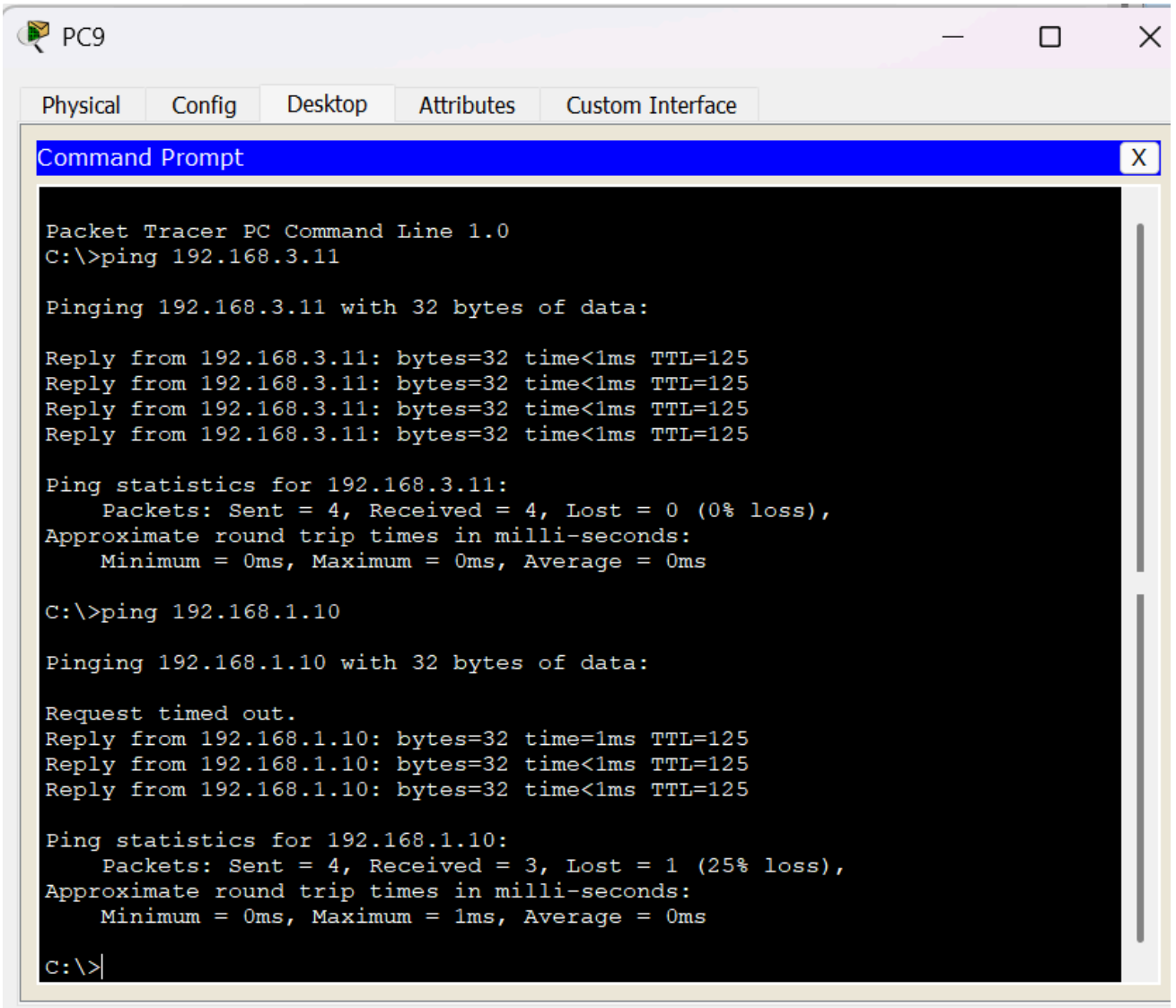
```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.5.11

Pinging 192.168.5.11 with 32 bytes of data:

Reply from 192.168.5.11: bytes=32 time<1ms TTL=124
Reply from 192.168.5.11: bytes=32 time<1ms TTL=124
Reply from 192.168.5.11: bytes=32 time<1ms TTL=124
Reply from 192.168.5.11: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.5.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

PC10

Physical | Config | Desktop | Attributes | Custom Interface

Command Prompt

```
Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=1ms TTL=126
Reply from 192.168.1.11: bytes=32 time<1ms TTL=126
Reply from 192.168.1.11: bytes=32 time<1ms TTL=126
Reply from 192.168.1.11: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.3.10

Pinging 192.168.3.10 with 32 bytes of data:

Reply from 192.168.3.10: bytes=32 time<1ms TTL=124
Reply from 192.168.3.10: bytes=32 time<1ms TTL=124
Reply from 192.168.3.10: bytes=32 time<1ms TTL=124
Reply from 192.168.3.10: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.3.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```



SWOT Analysis: Railway Track Signal Network Project

1) Strengths (Internal, Positive)

- **Safety:** Automated signals and track sensors prevent train collisions.
- **Reliability:** Ring topology ensures continuous communication even if a link fails.
- **Centralized Monitoring:** Star topology allows easy control from central PCs.
- **Scalability:** New PCs, switches, or track sections can be added easily.
- **Real-time Operation:** Train positions are updated instantly for effective signal control.

2)Weaknesses (Internal, Negative)

- **High Cost:** Routers, switches, and cabling are expensive.
- **Complex Setup:** Requires knowledge of networking, IP addressing, and routing.
- **Maintenance:** Regular checks needed for hardware, cables, and software.
- **Dependency on Network:** Failure in routers or switches can disrupt the system.

3) Opportunities (External, Positive)

- **Integration with IoT:** Sensors and signals can be upgraded to smart IoT devices.
- **Expansion:** Can be implemented on larger railway networks.
- **Automation:** Opportunity to integrate AI or predictive algorithms for train scheduling.
- **Real-world Application:** Modern railway systems are moving toward fully automated signaling.

4) Threats (External, Negative)

- **Power Failure:** System depends on uninterrupted electricity or UPS.
- **Cybersecurity Risks:** Networked signals can be vulnerable to hacking if not secured.
- **Environmental Factors:** Physical damage to cables or devices due to weather.
- **Obsolescence:** Newer technologies may replace current networking hardware over time.